CONNECTIONS ON PRINCIPAL FIBRE BUNDLES

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Abstract. Complete definitions and motivations are given for principal fibre bundles, connections on these, and associated vector bundles. The relationships between various concepts of differentiation in these settings are proved. Finally, we briefly give an outline of the theory of curvature in terms of connections.

1. Introduction

The motivation for defining *principal fibre bundles* and *connections* on them comes from the idea of *parallel transport* and *covariant differentiation* on a smooth manifold. Defining a derivative of one vector field with respect to another requires a rule for comparing vectors in nearby tangent spaces. The tangent space to each point of a manifold looks the same, that is, each is isomorphic to $\mathbb{R}^n$, but there is no canonical isomorphism between these tangent spaces. As such, there is no way to differentiate a vector field along a curve using the usual idea of the limit of a difference quotient, because the vectors being compared live in different tangent spaces. The solution to this problem is to define a rule for parallel transport on the tangent bundle (as a set, the collection of tangent spaces), which, given a smooth curve joining two points on a manifold, produces an isomorphism of the corresponding tangent spaces. This then enables vector fields to be differentiated along curves, and we think of the covariant derivative of one vector field with respect to another as the derivative of the first along an integral curve of the second.

We can further note that parallel transport can not only be thought of as carrying a single vector from one tangent space to another, but as carrying a frame for one to a frame for the other. Thus parallel transport can be thought of as taking place on the frame bundle (again, as a set at least, the collection of frames at each point on the manifold). Another way of thinking about the parallel transport is in terms of an *equivariant path lifting*. This is a rule that associates with each path in the manifold, and frame at the initial point of the path, a path in the frame bundle (that is, a frame at each point of the path). Equivariance here means that for a given curve in the manifold, the path starting at one frame is determined by the path starting at another frame because the first frame can be written in terms of the second. The parallel

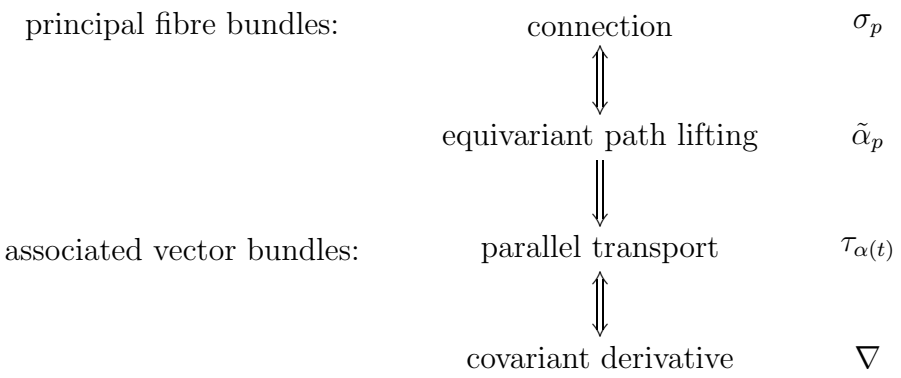
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transport is then the linear map from the frame at the initial point to the frame at the final point of the curve.

We can achieve this path lifting by specifying a rule which assigns to each vector at a point on the manifold, and frame at the point, a tangent vector to the frame bundle. We interpret this tangent vector as indicating how the frame must change as we move in the direction of the original vector on the manifold. This rule is called a *horizontal lift*, or *connection*. The lifted path is then defined as the integral curve of the horizontal lift of the tangent vectors to the curve in the manifold.

These ideas are formalised in the fairly abstract setting of vector bundles, principal fibre bundles and their structural groups, and associated bundles. We will define all of these ideas in the subsequent sections, but also make concrete reference to the more familiar tangent and frame bundles. The tangent bundle is a typical example of a vector bundle with $\mathbb{R}^n$ as a fibre, and the frame bundle is a typical principal fibre bundle, with $GL(n, \mathbb{R})$ as the structural group. The tangent bundle can additionally be described as an associated bundle of the frame bundle.

The connections between all these ideas are indicated in the following diagram.



1.1. **Notation used.** Throughout, M and P denote smooth manifolds. $C^\infty(M)$ denotes the algebra of smooth real valued functions on M. We will assume familiarity with the concept of a tangent bundle TM, including its smooth structure. $\mathfrak{X}(M)$ denotes the set of smooth vector fields on M. If G is a Lie group, $L(G)$ denotes its Lie algebra. $\mathrm{Ad}(g)$ is the adjoint representation of G on $L(G)$, defined as the derivative of the map $g' \mapsto gg'g^{-1}$. The Lie bracket of elements X, Y of $L(G)$ is denoted $[X, Y]$, and if (e_α) is a basis for $L(G)$, the structure coefficients are defined by $[e_\alpha, e_\beta] = C^\gamma_{\alpha\beta} e_\gamma$. If G acts on the right on some smooth manifold M, we write this action as $(p, g) \mapsto pg$. For fixed g, g_* and g^* denote the derivative of this map and the pull-back by this map respectively. For these and other constructions of basic differential geometry, refer to Boothy [1] or the more abstract but more

comprehensive Kobayashi and Nomizu [5] [6]. I and I' denote intervals in $\mathbb{R}$ containing 0.

2. The tangent bundle

If M is a smooth n-manifold, T_mM denotes the tangent space at the point m. T_mM can be thought of as the vector space of derivations at m. The tangent bundle TM is then the union of these tangent spaces:

$$TM = \{(m, X_m) | m \in M, X_m \in T_mM\}.$$

There is a natural projection map $\pi : TM \to M$ defined by $\pi(m, X_m) = m$. Notice that for each $m \in M$, $\pi^{-1}(m) = T_mM$, which is linearly isomorphic to $\mathbb{R}^n$. If $U \subset M$ is a local coordinate neighbourhood with coordinates $(x^1, \ldots, x^n)$ then $(e_i)_u = \frac{\partial}{\partial x^i} |_u$ constitute a basis for T_uU, and an element $X_u \in T_uU$ can be written $X_u = a^i(e_i)_u$. (x^i, a^i) then define a coordinate system for TU, and a covering of these coordinate systems defines the smooth structure on TM.

2.1. **The covariant derivative.** A covariant derivative is a mapping $\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$ denoted by $\nabla : (X, Y) \to \nabla_X Y$ which is $C^\infty(M)$-linear with respect to X, $\mathbb{R}$-linear with respect to Y, and satisfies the Leibniz rule

$$\nabla_X(fY) = (Xf)Y + f\nabla_X Y.$$

It is possible to prove (easily, using local representatives of connection forms, introduced later) that such a covariant derivative exists on any paracompact manifold. The covariant derivative from Riemannian geometry (as used in general relativity, for instance) is an example of this type of mapping. In that context further restrictions (being metric and torsion free) are placed on the connection, and it can be proved that there is a unique such connection for each Riemannian metric.

2.2. **Parallel transport.** A parallel transport on TM is a rule that assigns for each curve $\alpha : I \to M$ and $t \in I' \subset I$ a linear isomorphism $\tau_{\alpha(t)} : T_{\alpha(0)}M \to T_{\alpha(t)}M$ such that $t \mapsto \tau_{\alpha(t)}X_{\alpha(0)}$ is a smooth map from I' to TM for every $X_{\alpha(0)} \in T_{\alpha(0)}M$, and $\tau_{\alpha(t)}(AX_{\alpha(0)}) = A\tau_{\alpha(t)}X_{\alpha(0)}$ for every $A \in GL(n, \mathbb{R})$. This last property is called *equivariance*. In this context, it is perhaps best understood in terms of A representing a passive transformation which changes the basis. Equivariance then ensures that parallel transport is independent of the bases chosen in each of the two tangent spaces.

It can be shown that defining a parallel transport is equivalent to defining a covariant derivative, using

$$(\nabla_X Y)_m = \lim_{t \to 0} \frac{\tau^{-1}_{\alpha(t)} Y_{\alpha(t)} - Y_{\alpha(0)}}{t},$$

where α is the integral curve of X with $\alpha(0) = m$. Conversely, a parallel transport can be obtained from a covariant derivative by solving

the *parallel transport equation*, $\nabla_X Y = 0$, where X is a vector field extending $\dot{\alpha}$. The details of these proofs will be left until the general definition of a vector bundle has been given.

3. Vector bundles and principal fibre bundles

The definitions of this section are largely drawn from Isham [4]. The definition of a principal fibre bundle given here explicitly requires that the local trivialisations respect the group action.

3.1. **Locally trivial fibre bundles.** Our primary interest is in *locally trivial fibre bundles*, so we shall begin defining these.

A bundle $\xi = P \xrightarrow{\pi} M$ consists of a pair of smooth manifolds, P and M, respectively called the *total space* and the *base space*, and a map $\pi : P \to M$ called the projection map. We write $P(\xi) = P$ to denote the total space of the bundle ξ, and $M(\xi) = M$ to denote the base space.

A fibre bundle $\xi = P \xrightarrow{\pi} M$ with fibre F is a bundle such that for each $m \in M$, $\pi^{-1}(m)$ is diffeomorphic to F. We will at times denote the fibre over a point p by $P_p = \pi^{-1}(p)$. This partitioning of P into $\bigcup_{m \in M} \pi^{-1}(m)$ is referred to as the *fibration*.

A fibre bundle morphism from a fibre bundle $\xi = P \xrightarrow{\pi} M$ with fibre F to a fibre bundle $\eta = P' \xrightarrow{\pi'} M'$ with fibre F' is a pair of maps (ϕ, f) so $\phi : P \to P'$, $f : M \to M'$, and $\pi' \circ \phi = f \circ \pi$, so the following diagram commutes.

$$\begin{array}{ccc} P & \xrightarrow{\phi} & P' \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi'} \\ M & \xrightarrow{f} & M' \end{array}$$

This ensures that the maps respect the fibre structure.

Morphisms can be composed, as $(\phi', f') \circ (\phi, f) = (\phi' \circ \phi, f' \circ f)$. If $M = M'$, we say ξ and η are M-isomorphic, $\xi \cong \eta$, if there are morphisms $(\phi, f) : \xi \to \eta$, $(\phi', f') : \eta \to \xi$ so $(\phi', f') \circ (\phi, f) = (id_P, id_M)$ and $(\phi, f) \circ (\phi', f') = (id_{P'}, id_M)$. The bundle ξ is said to be *trivial* if it is M-isomorphic to $M \times F \xrightarrow{\pi} M$, the product fibre bundle.

We can also restrict bundles. If $N \subset M$, define

$$\xi \mid_N = \pi^{-1}(N) \xrightarrow{\pi|_{\pi^{-1}(N)}} N.$$

With this idea, we can say the bundles ξ and η are locally isomorphic if there is an open covering $\bigcup_\alpha U_\alpha$ of M so for each α, $\xi \mid_{U_\alpha}$ and $\eta \mid_{U_\alpha}$ are U_α-isomorphic. We can now define locally trivial as meaning locally isomorphic to the product fibre bundle. The open covering and the corresponding bundle morphisms are referred to as the *trivialisation* of the fibre bundle. There is not generally a distinguished trivialisation.

A *section* of a bundle is a smooth map $\sigma : M \to P$ such that $\pi \circ \sigma = id_M$. It assigns to each point $m \in M$ a point in the fibre of m. A local section is simply a section defined only on some open set of M. The set of sections of P is denoted $\Gamma P = \{\sigma : M \to P \mid \pi \circ \sigma = id_M\}$.

3.2. **Vector bundles.** A vector bundle is then a locally trivial fibre bundle with some vector space V as the fibre, so that the local trivialisations respect the linear structure. That is, if there is a local bundle isomorphism (ϕ, f) so $\phi : U \times F \to P(\xi \mid_U) = \pi^{-1}(U)$, then for every $u \in U$, $\phi : u \times V \to \pi^{-1}(u) \simeq V$ is a linear isomorphism.

3.3. **Principal fibre bundles.** If P is a smooth manifold and a Lie group G acts on P on the right, let P/G denote the set of orbits of the G-action, and $\rho : P \to P/G$ be the projection of a point onto its orbit. A locally trivial fibre bundle $\xi = P \xrightarrow{\pi} M$ is a principal fibre bundle with structural group G if there is a free right G-action on P so

- ξ is isomorphic to $P \xrightarrow{\rho} P/G$,
- the fibre is diffeomorphic to G,
- the local trivialisations respect the group action on P (this is made precise below).

Specifically, if there is a local trivialisation (ϕ, f) so $\phi : U \times G \to \pi^{-1}(U)$, then for every $u \in U$, $g \mapsto \phi(u, g) \in \pi^{-1}(u) \simeq G$ is a homomorphism. We write $\xi = G \rightsquigarrow P \xrightarrow{\pi} M$.

To be more explicit, we write the G-action on P as $(p, g) \mapsto pg$. Each fibre consists of the points in an orbit, and since the action is free, $pg = p$ implies $g = id$, so there is a bijection between each orbit and G. Since the action is smooth and invertible, each orbit must be diffeomorphic to G. Thus requirement that the fibre be diffeomorphic to G is redundant.

At times we will use the notation $\tau : P_m \times P_m \to G$ for the *translation function* of the principal fibre bundle, so $\tau(p', p)$ is the unique element of G so $p = p'\tau(p', p)$. We could think of this as "$\tau(p', p) = p'^{-1}p$".

We now prove the following result about local trivialisations.

Theorem 3.1. *Local sections are in one to one correspondence with local trivialisations*

Proof. Suppose $\sigma : U \subset M \to P$ is a local section. Then we can define a local trivialisation $\phi : U \times G \to \pi^{-1}(U)$ by $\phi(u, g) = \sigma(u)g$. It is easy to see that this is a diffeomorphism, with inverse $\phi^{-1}(p) = (\pi(p), \tau(\sigma(\pi(p)), p))$, and that it respects the group action. Conversely, if $\phi : U \times G \to \pi^{-1}(U)$, define a section $\sigma : U \to P$ by $\sigma(u) = \phi(u, id_G)$. □

4. Differentiation on vector bundles

Throughout this section, let $\xi = E \xrightarrow{\pi} M$ be a vector bundle with a vector space V as the fibre. We will denote the automorphisms of V by $\mathrm{Aut}(V)$. For instance, if $V = \mathbb{R}^n$, $\mathrm{Aut}(V) = GL(n, \mathbb{R})$.

4.1. **The covariant derivative.** We generalise the definition of a covariant derivative given in §2.1 to a general vector bundle.

A covariant derivative is a mapping $\nabla : \mathfrak{X}(M) \times \Gamma E \to \Gamma E$ denoted by $\nabla : (X, Y) \to \nabla_X Y$ which is $C^\infty(M)$-linear with respect to X, $\mathbb{R}$-linear with respect to Y, and satisfies the Leibniz rule

$$\nabla_X(fY) = (Xf)Y + f\nabla_X Y.$$

Notice that in the tangent bundle case this reduces to the earlier definition, because ΓTM is exactly what we mean by $\mathfrak{X}(M)$.

4.2. **Parallel transport.** Again, we give a simple generalisation of the definition of a parallel transport given in §2.2.

A parallel transport on E is a rule that assigns for each curve $\alpha : I \to M$ and $t \in I' \subset I$ a linear isomorphism $\tau_{\alpha(t)} : E_{\alpha(0)} \to E_{\alpha(t)}$ such that $t \mapsto \tau_{\alpha(t)} Y_{\alpha(0)}$ is a smooth map from I' to E for every $Y_{\alpha(0)} \in E_{\alpha(0)}$, and $\tau_{\alpha(t)}(AY_{\alpha(0)}) = A\tau_{\alpha(t)}Y_{\alpha(0)}$ for every $A \in \mathrm{Aut}(V)$.

4.3. **Equivalence of covariant derivatives and parallel transports.** We define a parallel transport using the *parallel transport equation* $\nabla_X Y = 0$. We will show that in the local coordinates induced by a vector bundle local trivialisation this equation reduces to an ordinary differential equation along the integral curves of X.

Theorem 4.1. *A covariant derivative can be used to define a parallel transport.*

Proof. Choose a point $m \in M$, and an open set U containing m so that U is a coordinate chart of M and E is locally trivial over U. Let $(x^1, \ldots, x^n)$ be local coordinates on U, and if $\phi : U \times V \to \pi^{-1}(U)$ is a local trivialisation, let $(\nu_1, \ldots, \nu_{n'})$ be a basis for V, and at each $u \in U$ define $(\mu_1, \ldots, \mu_{n'})$ where $\mu_i = \phi(u, \nu_i)$. This is a basis for $\pi^{-1}(u)$ since the local trivialisation respects the vector space structure. Further define $e_i = \frac{\partial}{\partial x^i}$ as a basis for $T_u U$. Define Γ^l_{ik} by $\nabla_{e_i}\mu_k = \Gamma^l_{ik}\mu_l$.

If $\alpha : I \to U$, $\alpha(0) = m$ has coordinates $(a^i(t))$, $\dot\alpha(t) = \dot a^i e_i = X$. Extend X to a vector field on U. If $Y_0 \in E_m$, we want to define $Y \in \Gamma E$ so $Y_m = Y_0$ and $\nabla_X Y = 0$ on $\alpha(I')$ for some possibly smaller interval

I'. Writing $Y(x^i) = (x^i, b^k\mu_k)$, we obtain

$$\begin{aligned}\nabla_X Y &= \nabla_{\dot{a}^i e_i} b^k \mu_k \\ &= \dot{a}^i \nabla_{e_i} b^k \mu_k \\ &= \dot{a}^i (\frac{\partial b^k}{\partial x^i}\mu_k + b^k \Gamma^l_{ik}\mu_l) \\ &= \dot{a}^i (\frac{\partial b^l}{\partial x^i} + b^k\Gamma^l_{ik})\mu_l.\end{aligned}$$

Setting this to zero, we have $\frac{d}{dt}(b^l \circ \alpha) + (b^k \circ \alpha)\dot{a}^i(\Gamma^l_{ik} \circ \alpha) = 0$. These constitute n' first order ordinary differential equations in t, and so given the initial condition that $b^l(0) = Y^l_0$ there is a unique solution $b^l(t)$ on some interval I' containing 0. We then define $\tau_{\alpha(t)}Y_0 = (\alpha(t), b^l(t)\mu_l)$. Since these equations are linear, the equivariance property of the parallel transport is assured. □

Conversely we define a covariant derivative as follows. If $X \in \mathfrak{X}(M)$, let $\alpha : I \to M$ be an integral curve of X so $\alpha(0) = m$. Define

$$(\nabla_X Y)_m = \lim_{t\to 0} \frac{1}{t}(\tau^{-1}_{\alpha(t)} Y(\alpha(t)) - Y(\alpha(0))).$$

However, with the definition of a parallel transport we have given, we cannot ensure that this recipe for a covariant derivative works. In particular, we cannot yet prove the $C^\infty(M)$-linearity with respect to X. We will see later that vector bundles with fibre V can be thought of as associated bundles of principal fibre bundles with structural group $\mathrm{Aut}(V)$, and that a connection defined on this principal fibre bundle induces a parallel transport on the vector bundle. Using a parallel transport induced in this way, we will be able to complete the proof that the above recipe defines a covariant derivative. The reason why we are unable to do this at present is that further conditions are needed specifying how the parallel transport varies when the curve is varied. Certainly choosing a different curve between the same two points can lead to a different parallel transport (for example, parallel transport around most closed loops on the sphere is different from parallel transport along the constant loop!). However, we would expect that reparametrising the curve should not change the parallel transport, and we have not enforced this condition.

Theorem 4.2. *The formula for a covariant derivative in terms of a parallel transport is $\mathbb{R}$-linear in Y, and satisfies the Leibniz rule.*

Proof. We will look at this equation in local coordinates. As above, let (x^i) denote coordinates on $U \subset M$, $e_i = \frac{\partial}{\partial x^i}$ form a basis for T_uU, and μ_j be a basis for E over $\pi^{-1}(U)$. Now $\tau^{-1}_{\alpha(t)} : E_{\alpha(t)} \to E_{\alpha(0)}$, so define $b^k_j(t)$ by $\tau^{-1}_{\alpha(t)}\mu_j = b^k_j(t)\mu_k$. By the equivariance property of the parallel transport, if $Y(\alpha(t)) = v^j(t)\mu_j$ then $\tau^{-1}_{\alpha(t)}Y(\alpha(t)) = v^j(t)b^k_j(t)\mu_k$. Note

$b_j^k(0) = \delta_j^k$. Thus

$$(\nabla_X Y)_m = \lim_{t\to 0} \frac{1}{t}(v^j(t)b_j^k(t) - v^k(t))\mu_k$$

so in component form

$$\begin{aligned}(\nabla_X Y)_m^k &= \lim_{t\to 0}\frac{1}{t}(v^j(t)b_j^k(t) - v^k(t)b_j^k(0)) \\ &= \frac{d}{dt}(b_j^k(t)v^j(t)) \mid_{t=0} \\ &= v^j(0)\frac{d}{dt}b_j^k(t) \mid_{t=0} + \frac{d}{dt}v^k(t) \mid_{t=0} \\ &= (Xv^k)_m + v^j(0)(Xb_j^k)_m.\end{aligned}$$

We have assumed here that v_k and b_j^k can be extended to functions on an open set containing $\alpha(0)$. $\mathbb{R}$-linearity in Y is obvious from this expression. The Leibniz rule is seen in the following calculation. Let $f \in C^\infty(M)$.

$$\begin{aligned}(\nabla_X fY)_m^k &= (Xf)_m v^k(0) + f(m)(Xv^k)_m + f(m)v^j(0)(Xb_j^k)_m \\ &= (Xf)_m v^k(0) + f(m)(\nabla_X Y)_m^k.\end{aligned}$$

We can see why proving $C^\infty(M)$-linearity in X is impossible at this point. Specifically, changing X results in the integral curve α changing also, resulting in new coefficients b_j^k describing the parallel transport. As we have not given any condition on how the parallel transport must depend on the curve, we cannot control the change in b_j^k. We will return to this theorem in §6.1 and show that when the parallel transport is induced by a connection on the principal fibre bundle which the vector bundle is associated to, there are conditions on the parallel transport which allow us to prove that the covariant derivative defined here is in fact correct. □

5. Principal fibre bundles

5.1. **Connections on principal fibre bundles.** There are a number of equivalent ways of defining a *connection* on a principal fibre bundle. We give two closely related definitions here, following [2]. Throughout $\xi = G \rightsquigarrow P \xrightarrow{\pi} M$ is a principal fibre bundle. We first define the *vertical subspace* at a point p of P. $V_p = \{X \in T_pP \mid \pi_* X = 0\}$. The vertical vectors can be thought of as those which point only along the fibre.

Definition 5.1 (Horizontal lift)**.** *A connection is a smooth choice for each $p \in P$ of a linear map $\sigma_p : T_mM \to T_pP$ where $m = \pi(p)$ so that*

- $\pi_* \circ \sigma_p = id_{T_mM}$,
- $\sigma_{pg} = g_*\sigma_p$ *(the 'elevator' property).*

Definition 5.2 (Horizontal subspace)**.** *A connection is a smooth choice for each $p \in P$ of a subspace H_p of T_pP so that*

- $T_pP = V_p \oplus H_p$,
- $H_{pg} = g_* H_p$ *(the 'elevator' property).*

These are equivalent, as follows. Given a choice of a horizontal lift, for each $p \in P$, define $H_p = \sigma_p(T_m M)$. Then if $X_p \in H_p$, we cannot have $\pi_* X_p = 0$ except when $X_p = 0$, since $\pi_* \circ \sigma_p = id_{T_m M}$. Further, for any $X_p \in H_p$, we can write $X_p = \sigma_p \pi_* X_p + (X_p - \sigma_p \pi_* X_p)$, and $\sigma_p \pi_* X_p \in H_p$, and $\pi_*(X_p - \sigma_p \pi_* X_p) = \pi_* X_p - \pi_* X_p = 0$ so $X_p - \sigma_p \pi_* X_p \in V_p$. Thus $T_pP = V_p \oplus H_p$, and the elevator property for H_p follows from the elevator property of σ_p, so the subspaces defined in this way constitute a connection. Conversely, given a choice of horizontal subspaces, if $X_p \in H_p$, and $X_p \neq 0$, then $\pi_* X_p \neq 0$, so $\pi_* \mid_{H_p}$ is injective. Suppose (ϕ, f) is a local trivialisation so $\phi : \pi^{-1}(U) \to U \times G$. If $\kappa : U \times G \to U$ is the projection, $\pi = \kappa \circ \phi$, so $\pi_* = \kappa_* \circ \phi_*$, and since κ_* is clearly surjective, and ϕ is a diffeomorphism, π_* is surjective also. Thus $\pi_* \mid_{H_p}$ must be surjective, since $V_p = \ker \pi_*$. Since $\pi_* \mid_{H_p}$ is a linear isomorphism, we can define $\sigma_p = \pi_* \mid_{H_p}^{-1}$. Now $\sigma_{pg}^{-1} = \pi_* \mid_{g_* H_p}$ and $(g_* \sigma_p)^{-1} = \sigma_p^{-1} g_*^{-1} = \pi_* \mid_{H_p} (g_*^{-1}) = \pi_* \mid_{g_* H_p}$ since $\pi \circ g = \pi$. Thus the elevator property holds, and the σ_p define a connection in terms of horizontal lifts.

We define the horizontal and vertical components of a vector $X \in T_pP$, hor$X$ and ver$X$ as the components in the horizontal and vertical subspaces respectively. Alternatively, hor$X = \sigma_p \pi_* X$, and ver$X = X - \sigma_p \pi_* X$.

We say a curve $\gamma : I \to P$ is horizontal if $\dot{\gamma}(t) \in H_{\gamma}(t)$ for all $t \in I$.

5.2. **Equivariant path liftings.** Equivariant path lifting can be thought of as a type of parallel transport, in the setting of a principal fibre bundle. An equivariant path lifting associates with each smooth curve $\alpha : I \to M$ and $p \in \pi^{-1}(\alpha(0))$ a curve $\tilde{\alpha}_p : I' \to P$ with $I' \subset I$ so $\pi \circ \tilde{\alpha}_p = \alpha$, and $\tilde{\alpha}_{pg}(t) = \tilde{\alpha}_p(t) g$ for all $g \in G$ and $t \in I'$.

Theorem 5.3. *A connection defines an equivariant path lifting.*

Proof. We will only prove that a connection can be used to define equivariant path liftings of simple curves. The following proof fills in the details of the sketch in [7]. See [4] for a more general proof.

Choose $\alpha : I \to M$ and $p \in \pi^{-1}(\alpha(0))$. Define $X \in \mathfrak{X}(\pi^{-1}(\alpha(I)))$ by $X_q = \sigma_q \dot{\alpha}(t)$ where $\alpha(t) = \pi(q)$. Now extend X to an open set in P containing $\pi^{-1}(\alpha(I))$. Let $\tilde{\alpha}_p$ be the unique integral curve of X defined on some interval $I' \subset I$ so that $\tilde{\alpha}_p(0) = p$.

Let $J = \{t \in I' \mid \pi(\tilde{\alpha}_p(t)) = \alpha(t)\}$. J is clearly closed, since both expressions in the equation are continuous in t, and J is non-empty, since $\pi(\tilde{\alpha}_p(0)) = \alpha(0)$. Now suppose that $\pi(\tilde{\alpha}_p(t_0)) = \alpha(t_0)$ for some

$t_0 \in I'$. Now

$$\begin{aligned}\frac{d}{dt}(\pi(\tilde{\alpha}_p(t)) &= \pi_* \dot{\tilde{\alpha}}_p(t) \\ &= \pi_* X_{\tilde{\alpha}_p(t)} \\ &= \pi_* \sigma_{\tilde{\alpha}_p(t)} \dot{\alpha}(t) \\ &= \dot{\alpha}(t) = \frac{d}{dt}\alpha(t).\end{aligned}$$

Thus by the uniqueness theorem for solutions of ordinary differential equations $\pi(\tilde{\alpha}_p(t)) = \alpha(t)$ for all t in some open set around t_0. Thus J is also open, and since I' is connected, $J = I'$, so $\pi \circ \tilde{\alpha}_p = \alpha$.

Next let $K_g = \{t \in I' \mid \tilde{\alpha}_{pg}(t) = \tilde{\alpha}_p(t)g\}$. Again, K_g must be closed, and $\tilde{\alpha}_{pg}(0) = pg = \tilde{\alpha}_p(t)g$ so $0 \in K_g$. Suppose for some $t_0 \in I'$ $\tilde{\alpha}_{pg}(t_0) = \tilde{\alpha}_p(t_0)g$. Now

$$\begin{aligned}\frac{d}{dt}\tilde{\alpha}_{pg}(t) &= X_{\tilde{\alpha}_{pg}(t)} \\ &= \sigma_{\tilde{\alpha}_{pg}(t)} \dot{\alpha}(t) \\ &= \sigma_{\tilde{\alpha}_p(t)g} \dot{\alpha}(t) \\ &= g_* \sigma_{\tilde{\alpha}_p(t)} \dot{\alpha}(t) \\ &= g_* \dot{\tilde{\alpha}}_p(t) \\ &= \frac{d}{dt}(\tilde{\alpha}_p(t)g),\end{aligned}$$

so by the uniqueness theorem for solutions of ordinary differential equations $\tilde{\alpha}_{pg}(t) = \tilde{\alpha}_p(t)g$ for all t in some open set around t_0. Thus K_g is open, and so $K_g = I'$, and $\tilde{\alpha}_{pg}(t) = \tilde{\alpha}_p(t)g$ for all $g \in G$ and $t \in I'$, completing the proof. □

Notice that the proof of this theorem gives an additional property of the equivariant path lifting, namely that it lifts curves to horizontal curves in the total space.

Conversely, we can prove that an equivariant path lifting can be used to define a connection.

Theorem 5.4. *An equivariant path lifting defines a connection*

Proof. Choose a set of vector fields $(X^1, \ldots, X^n)$ so $(X^1_m, \ldots, X^n_m)$ spans $T_m M$, and let $(\alpha^1, \ldots, \alpha^n)$ be the integral curves of these vector fields. Given $p \in P$, define $\sigma_p X^i_m = \dot{\tilde{\alpha}}^i_p(0)$, where $\tilde{\alpha}^i$ is the equivariant path lifting of α^i starting at p. Extend σ_p linearly to all of $T_m M$.

Checking the two required properties of σ_p is simple.

$$\begin{aligned}\pi_*\sigma_p X^i &= \pi_*\dot{\tilde{\alpha}}^i_p(0)\\ &= \frac{d}{dt}(\pi(\tilde{\alpha}^i_p(t)))\mid_{t=0}\\ &= \frac{d}{dt}\alpha^i(t)\mid_{t=0}\\ &= X^i\end{aligned}$$

and

$$\begin{aligned}\sigma_{pg}X^i &= \dot{\tilde{\alpha}}^i_{pg}(0)\\ &= \frac{d}{dt}(\tilde{\alpha}^i_p(t)g)\mid_{t=0}\\ &= g_*\dot{\tilde{\alpha}}^i_p(0)\\ &= g_*\sigma_p X^i,\end{aligned}$$

and so these identities hold on all of T_mM. □

It is not hard to see that the lifted path is horizontal with respect to this connection, and that these two processes are inverses.

6. Associated bundles

We know reach the idea of an associated bundle. These can be thought of as derived from a principal fibre bundle by means of replacing the fibre G with another space having a left G-action. The definition given here is that from Isham [4].

If X and Y are spaces with a right and left G-action respectively, we define an equivalence relation $\sim$ on $X \times Y$ by $(x, y) \sim (xg, g^{-1}y)$ for all $g \in G$. The G-product $X \times_G Y$ is then defined to be $(X \times Y)/\sim$.

If $\xi = G \rightsquigarrow P \xrightarrow{\pi} M$ is a principal fibre bundle and there is a left G action on F then we define the fibre bundle *associated with ξ by the G-action on F* to be $\xi[F] = P_F \xrightarrow{\pi_F} M$, where $P_F = P \times_G F$ and $\pi_F([p, v]) = \pi(p)$.

π_F is well defined since if $(p, v) \sim (p', v')$, then there is a $g \in G$ so $p' = pg$, $v' = g^{-1}v$, so $\pi_F([p', v']) = \pi(p') = \pi(pg) = \pi(p) = \pi_F([p, v])$.

Notice that there is an obvious, but trivial, action of G on P_F, by $[p, v]g = [pg, g^{-1}v] = [p, v]$.

It is necessary to prove the $\xi[F]$ is in fact a fibre bundle, and that the fibre is F.

Theorem 6.1. *For each $m \in M$, $\pi_F^{-1}(m)$ is diffeomorphic to F, and so $\xi[F]$ is a fibre bundle with fibre F.*

Proof. Fix $q \in P_m$ and define $i_q : F \to P_F$ by $i_q(v) = [q, v]$. Since $\pi_F([q, v]) = m$, $i_q(F) \subset \pi_F^{-1}(m)$. Thus $i_q : F \to \pi_F^{-1}(m)$. Next define $j_q : \pi_F^{-1}(m) \to F$ by $j_q([p, v]) = \tau(q, p)v$ where τ is the translation

function described in §3.3. $j_q([pg, g^{-1}v]) = \tau(q, pg)g^{-1}v = \tau(q, p)v = j_q([p, v])$ so j_q is well-defined. Now $i_q \circ j_q([p, v]) = [q, \tau(q, p)v] = [q\tau(q, p), v] = [p, v]$ and $j_q \circ i_q(v) = \tau(q, q)v = v$, so i_q and j_q are diffeomorphisms. □

It can be proved that if $\xi = \mathrm{Aut}(V) \rightsquigarrow P \xrightarrow{\pi} M$ is a principal fibre bundle, then the associated bundle $\xi[V] = P_V \xrightarrow{\pi_V} M$ with $\mathrm{Aut}(V)$ acting on V as normal is in fact a vector bundle. Moreover, any vector bundle with fibre V is associated to the principal fibre bundle with structural group $\mathrm{Aut}(V)$.

6.1. **Parallel transports from equivariant path liftings.** We are now in a position to define a parallel transport on an associated vector bundle $\xi[V]$ given an equivariant path lifting on a principal fibre bundle ξ with structural group $\mathrm{Aut}(V)$. This completes the chain of relationships from connection through path lifting and parallel transport to covariant derivative.

We first define a path lifting as follows. If $\alpha : I \to M$ is a curve in M, and $\pi_V([p, v]) = \pi(p) = \alpha(0)$, let $\tilde{\alpha}_p$ be the equivariant path lifting of α to the principal fibre bundle ξ. Then define a parallel transport on the associated vector bundle $\xi[V]$ as $\tau_{\alpha(t)} : \pi_V^{-1}(\alpha(0)) \to \pi_V^{-1}(\alpha(t))$ by $\tau_{\alpha(t)}([p, v]) = [\tilde{\alpha}_p(t), v]$. The equivariance of the path lifting with respect to $\mathrm{Aut}(V)$ guarantees that this definition is well-defined. It is also a parallel transport in the sense of §4.2, because the G-action on P_F is in fact trivial, as described previously.

We can now briefly finish the proof of §4.3, showing that when a parallel transport is induced in this fashion from the principal fibre bundle, the covariant derivative $\nabla_X Y$ defined in terms of the parallel transport is $C^\infty(M)$-linear in X. Consider a local trivialisation over U of both the vector bundle and the principal fibre bundle. Choose bases (e_k) for TU, (μ_j) for V, and (ε_i^j) for $L(\mathrm{Aut}(V))$. Define $\sigma_{n\,j}^{\ i}(p)$ by $\sigma_p e_n = \sigma_{n\,j}^{\ i}(p)\varepsilon_i^j$. Then if $\dot{\alpha}(t) = q^n(t)e_n$, $\sigma_{\alpha(t)}\dot{\alpha}(t) = q^n(t)\sigma_{n\,j}^{\ i}(\alpha(t))\varepsilon_i^j$. Let $\tilde{\alpha}$ denote the equivariant path lifting of α which at $\alpha(0)$ gives the identity in $\mathrm{Aut}(V)$. For convenience, write $(\tilde{\alpha}(t))_j^k = g_j^k(t)$. Now $\dot{\tilde{\alpha}}(t) = \sigma_{\alpha(t)}\dot{\alpha}(t) = q^n(t)\sigma_{n\,j}^{\ k}(\alpha(t))\varepsilon_k^j$. Writing $Y(\alpha(t)) = v^j(t)\mu_j$, with this notation we obtain $\tau_{\alpha(t)}Y(\alpha(0)) = g_j^k(t)v^j(0)\mu_k$ or

$$\tau_{\alpha(t)}^{-1}Y(\alpha(t)) = (g^{-1})_j^k v^j(t)\mu_k.$$

Thus, following on from Theorem 4.2 in component form

$$\begin{aligned}
(\nabla_X Y)^k_m &= \lim_{t\to 0}\frac{1}{t}(v^j(t)g^k_j(t)^{-1} - v^k(t)g^k_j(0)^{-1}) \\
&= \frac{d}{dt}(g^k_j(t)^{-1}v^j(t)) \mid_{t=0} \\
&= v^j(0)\frac{d}{dt}g^k_j(t)^{-1} \mid_{t=0} + \frac{d}{dt}v^k(t) \mid_{t=0} \\
&= -v^j(0)g^k_j(0)^{-2}\frac{d}{dt}g^k_j(t) \mid_{t=0} + \frac{d}{dt}v^k(t) \mid_{t=0} \\
&= -v^j(0)q^n(0)\sigma_{nj}{}^k(0) + (Xv^k)_{\alpha(0)}
\end{aligned}$$

This expression is now clearly $C^\infty(M)$-linear in X, as $\sigma_{nj}{}^k$ is independent of X, and $q^n \propto X$.

6.2. **The tangent bundle as an associated bundle of the frame bundle.** If M is a smooth n-manifold, a frame f at $m \in M$ is a basis $(f_1, \ldots, f_n)$ for T_mM. The set of frames at m is denoted F_mM. The space of all frames on M, denoted by FM has a free right $GL(n,\mathbb{R})$-action, given by

$$fg = (f_1, \ldots, f_n)g = (g^i_1 f_i, \ldots, g^i_n f_i)$$

for $g \in GL(n,\mathbb{R})$. If $U \subset M$ is a local coordinate neighbourhood with coordinates $(x^1, \ldots, x^n)$ then $e_i = \frac{\partial}{\partial x^i} \mid_u$ constitute a basis for T_uU. An element $f \in F_uU$ can be written $f_j = f^k_j e_k$, where (f^k_j) are the components of a matrix in $GL(n,\mathbb{R})$. (x^i, f^k_j) then define a coordinate system for FU, and a covering of these coordinate systems defines the smooth structure on FM. Clearly FM has dimension $n + n^2$. With the obvious projection map $GL(n,\mathbb{R}) \rightsquigarrow FM \xrightarrow{\pi} M$ is a principal fibre bundle. A local section of the frame bundle is a choice of a local field of frames.

The tangent bundle is then formed as the associated bundle $FM[\mathbb{R}^n]$ with $GL(n,\mathbb{R})$ acting on $\mathbb{R}^n$ as usual. This associated bundle is identified with the tangent bundle by taking the point $[f, v]$ in $FM \times_{GL(n,\mathbb{R})} \mathbb{R}^n$ to $f_i v^i \in T_{\pi(f)}M$. This is well defined, since $[f, v] \sim [fg, g^{-1}v]$ which is taken to $f_j(g)^j_i(g^{-1})^i_k v^k = f_j \delta^j_k v_k = f_i v^i$.

7. Connection forms

This section describes the relationship between the previously given outline of the theory of connections on principal fibre bundles and the idea of curvature. For the sake of brevity, where results appear in standard texts, we refer to these for proofs.

7.1. **Definition of a connection form.** Alongside the idea of a connection as a horizontal lift or a horizontal subspace, we can think of a connection in terms of a *connection form*. This is a one form defined on the total space of a principal fibre bundle, with values in $L(G)$, the Lie algebra of the structural group.

Firstly, we define two maps identifying V_p with $L(G)$. Let $\theta_p : G \to P_p$ be given be $\theta_p(g) = pg$, and $\psi_p : P_p \to G$ be given by $\psi_p(p') = \tau(p, p')$. (τ is the translation function, described in §3.3.) Now $\theta_p(\psi_p(p')) = p'$ and $\psi_p(\theta_p(g)) = g$. Also $\pi \circ \theta_p(g) = \pi(pg) = p$ so $\pi \circ \theta_p$ is a constant function for each p, so $\pi_* \theta_{p*} v = (\pi \circ \theta_p)_* v = 0$, so $\theta_{p*} : L(G) \to V_p$ and $\psi_{p*} : V_p \to L(G)$ are linear isomorphisms.

We can now define the connection form ω, as $\omega_p(u) = \psi_{p*}(u)$ if $u \in V_p$ and $\omega_p(u) = 0$ if $u \in H_p$. Alternatively, $\omega_p(u) = \psi_{p*}(\mathrm{ver} u) = \psi_{p*}(u - \sigma_p \pi_* u)$. Thus the connection form annihilates the horizontal vectors, and takes the vertical vectors into the Lie algebra.

The connection form has a property corresponding to the 'elevator property' of σ_p.

Theorem 7.1. $(g^*\omega)_p(u) = \omega_{pg}(g_* u) = Ad(g^{-1})(\omega_p(u))$

Proof.

$$\begin{aligned} \omega_{pg}(g_* u) &= \psi_{pg*}(g_* u - \sigma_{pg} \pi_* g_* u) \\ &= \psi_{pg*}(g_* u - \sigma_{pg} \pi_* u) \\ &= \psi_{pg*}(g_* u - g_* \sigma_p \pi_* u) \\ &= (\psi_{pg} \circ g)_* (u - \sigma_p \pi_* u). \end{aligned}$$

Now $(\psi_{pg} \circ g)(p') = \psi_{pg}(p'g) = g^{-1}\tau(p, p')g = g^{-1}\psi_p(p')g$ so $(\psi_{pg} \circ g)_* = \mathrm{Ad}(g^{-1})\psi_{p*}$. Thus

$$\omega_{pg}(g_* u) = \mathrm{Ad}(g^{-1})(\psi_{p*}(u - \sigma_p \pi_* u)),$$

establishing the result. □

We can also define a connection form as a Lie algebra valued form on the total space, satisfying this property, and with $\omega_p(u) = \psi_{p*}(u)$ for all $u \in V_p$. Then there is a unique connection, defined by $H_p = \ker \omega_p$. It can be checked that the elevator property for ω_p implies that for H_p.

7.2. **Local representatives of the connection form.** The connection form we have defined here is a form on the total space. In many cases we will want to think of the connection in terms of a form defined on the base space. This is achieved through *local representatives*. Diverse examples relying on the idea of local representatives of forms on the total space include the matrix of $\mathbb{R}$-valued connection forms used in Riemannian geometry, the curvature of a Riemannian manifold in terms of local coordinates, or even the electromagnetic vector potential in a given gauge.

To obtain a form on the base space, we need some map from the base space to the total space, so that the connection form can be pulled back via this map. To this end, we take a local section $\sigma : U \subset M \to P$, and define the local representative of ω under σ to be $\omega^U = \sigma^*\omega$. In the context of a frame bundle, choosing a local section corresponds to choose a local frame field, and hence the local representative can be thought of as "the connection in local coordinates". When G is a real Lie group we can obtain a collection of $\mathbb{R}$-valued forms by choosing a basis for $L(G)$, and writing the $L(G)$-valued local representative in terms of this basis.

An extremely important result is that this process of forming a local representative can be reversed. Knowing the section used to define it, we can return to the original connection on the total space. This result is established using the local trivialisation corresponding to the local section. Define $\psi : U \times G \to \pi^{-1}(U)$ by $\psi(m, g) = \sigma(m)g$. As we have seen earlier, this is a diffeomorphism.

Theorem 7.2. *If $(\alpha, \beta) \in T_mU \oplus T_gG$ then*

$$(\psi^*\omega)_{(m,g)}(\alpha, \beta) = Ad(g^{-1})(\omega^U_m(\alpha)) + \beta$$

where $\beta \in T_gG$ is also understood as an element of the Lie algebra.

The proof is straightforward, and given in Isham [4]. Since ψ is a diffeomorphism, the original connection form on P can be obtained from this equation. This result makes the statement that the local representative is the connection form "in local coordinates" precise.

7.3. **Covariant derivative of $L(G)$-valued forms.** Given a $L(G)$-valued r-form φ on the total space P of a principal fibre bundle ξ, we can define its *covariant derivative* by

$$D\varphi(v_1, \dots, v_{r+1}) = d\varphi(\mathrm{hor}v_1, \dots, \mathrm{hor}v_{r+1}).$$

$\Omega = D\omega$ is called the *curvature form* of the connection ω.

Theorem 7.3 (Maurer-Cartan structural equations)**.** *If (e_α) is a basis for $L(G)$, then*

$$\Omega(X, Y) = d\omega(X, Y) + [\omega(X), \omega(Y)]$$

and in local coordinates, writing $\omega = \omega^\alpha e_\alpha$ and $\Omega = \Omega^\alpha e_\alpha$

$$\Omega^\alpha = d\omega^\alpha + \frac{1}{2}C^\alpha_{\beta\gamma}\omega^\beta \wedge \omega^\gamma.$$

Proof. For the first part, refer to either [4] or [2]. In local coordinates,

$$\begin{aligned}
[\omega(X),\omega(Y)] &= \omega(X)\omega(Y) - \omega(Y)\omega(X) \\
&= \omega^\alpha(X)e_\alpha\omega^\beta(Y)e_\beta - \omega^\beta(X)e_\beta\omega^\alpha(Y)e_\alpha \\
&= \omega^\alpha(X)\omega^\beta(Y)[e_\alpha, e_\beta] \\
&= \omega^\alpha(X)\omega^\beta(Y)C^\gamma_{\alpha\beta}e_\gamma \\
&= \frac{1}{2}C^\gamma_{\alpha\beta}e_\gamma(\omega^\alpha \wedge \omega^\beta)(X,Y),
\end{aligned}$$

establishing the second part. □

This results allows a simple proof of the following result.

Theorem 7.4 (Bianchi identity)**.** $D^2\omega = 0$.

Proof. $d\Omega^\gamma = \frac{1}{2}C^\gamma_{\alpha\beta}(d\omega^\alpha \wedge \omega^\beta - \omega^\alpha \wedge d\omega^\beta)$. Now

$$D\Omega^\gamma(X,Y) = d\Omega^\gamma(\mathrm{hor}X, \mathrm{hor}Y),$$

and $\omega^\beta(\mathrm{hor}Y) = \omega^\alpha(\mathrm{hor}X) = 0$, so $D\Omega = 0$. □

We can also form a local representative of the curvature form, by pulling it back by a section. Say $\sigma : U \to P$ is a section, and $A = \sigma^*\omega$ is the local representative of the connection. We will write $F = \sigma^*\Omega$ as the local representative of the curvature form. It is easy to see that $F^\gamma = dA^\gamma + \frac{1}{2}C^\gamma_{\alpha\beta}A^\alpha \wedge A^\beta$. The notation is intentionally suggestive of a relationship with electromagnetism — the local representative of the connection being the vector potential in a given gauge, and the local representative of the curvature being the Faraday tensor in this gauge.

7.4. **Covariant exterior derivative on an associated vector bundle.** We now specialise to the case of a $\mathrm{Aut}(V)$ principal fibre bundle $\xi = \mathrm{Aut}(V) \rightsquigarrow P \xrightarrow{\pi} M$, and define a *covariant exterior derivative* which acts on forms defined on the base space M with values in the associated vector bundle $\xi[V]$. This is a map

$$\mathbf{d} : \Omega^p(M, \xi[V]) \to \Omega^{p+1}(M, \xi[V])$$

satisfying the Leibniz rule, that is, if given forms $\alpha \in \Omega^p(M, \xi[V])$ and $\beta \in \Omega^q(M, \xi[V])$ then

$$\mathbf{d}(\alpha \wedge \beta) = (\mathbf{d}\alpha) \wedge \beta + (-1)^p\alpha \wedge (\mathbf{d}\beta).$$

Notice that we distinguish between $\mathbf{d}$, a map on vector bundle valued forms, and d, which is the usual exterior derivative acting on real valued forms. There are a number of alternative approaches to defining this map. In Darling [3] it is defined on $\Omega^0(M, \xi[V])$ in terms of the covariant derivative of §4.1 as the unique map so that for a section $\sigma \in \Gamma\xi[v] = \Omega^0(M, \xi[V])$ and vector field $X \in \mathfrak{X}(M)$

$$(\mathbf{d}\sigma)X = \nabla_X\sigma.$$

It is then extended to p-forms as follows. A p-form $\alpha \in \Omega^p(M, \xi[v])$ can be written as $\alpha = \alpha^j \sigma_j$. Here α^j is a $\mathbb{R}$-valued p-form, and the σ_j are $\xi[V]$-valued 0-forms, induced from a local section σ of the principal fibre bundle by $\sigma_j = [\sigma, \mu_j]$, where μ_j are a basis for V. Then

$$\mathbf{d}\alpha = (d\alpha^j)\sigma_j + (-1)^p \alpha^j (\mathbf{d}\sigma_j)$$

This will guarantee that the exterior derivative satisfies the Leibniz rule. For a proof that this is the unique such map, see [3].

We prefer to define the exterior derivative in terms of the connection form. It will not be an intrinsic definition, as it depends upon a choice of a local section $\sigma : U \to P$. Given such a local section, we work in the local trivialisation defined by it. Let (e_j) be a basis for V. Let (ε_i^j) be the canonical basis for $L(\mathrm{Aut}(V))$ in this local trivialisation. The local section gives a local basis for $\xi[V]$, denoted by $\sigma_j = [\sigma, e_j]$ Write the connection form ω in terms of $\mathbb{R}$-valued 1-forms, so $\omega = \omega_j^i \varepsilon_i^j$, and $A = \sigma^* \omega$, $A_j^i = \sigma^* \omega_j^i$. We now begin defining the exterior derivative. Firstly, on the basis σ_j, $\mathbf{d}\sigma_j = \sigma_i A_j^i$. Then any section s of $\xi[V]$ can be written $s = s^j \sigma_j$, where $s^j \in C^\infty(M)$. Thus we define $\mathbf{d}s = (ds^j)\sigma_j + s^j \sigma_i A_j^i$. Finally we extend $\mathbf{d}$ to p-forms as above, so that it satisfies the Leibniz rule. It is necessary to check that given a different local section we obtain the same exterior derivative. This will rely largely on the elevator property of the connection form. This proof is omitted. Further, once this invariance has been established, since Darling's [3] definition of the exterior derivative agrees with ours in local coordinates, we can use that definition, in terms of the covariant derivative on the vector bundle, to give another route from a connection on a principal fibre bundle to a covariant derivative on an associated vector bundle.

7.5. **The second exterior derivative, curvature, and the Riemann tensor.** Using the above definitions, given a local section of the principal fibre bundle, we can calculate the second exterior derivative of any $\xi[V]$-valued form. Say $\alpha \in \Omega^p(M, \xi[v])$ is given in terms of the basis induced from the section as $\alpha = \sigma_j \alpha^j$. Then

$$\begin{aligned} \mathbf{d}\alpha &= (\mathbf{d}\sigma_j)\alpha^j + \sigma_j (d\alpha^j) \\ &= \sigma_i A_j^i \wedge \alpha^j + \sigma_j (d\alpha^j). \end{aligned}$$

Next,

$$\begin{aligned} \mathbf{dd}\alpha &= (\mathbf{d}\sigma_i) \wedge A_j^i \wedge \alpha^j + \sigma_i (dA_j^i) \wedge \alpha^j - (\sigma_i A_j^i) \wedge (d\alpha^j) + \\ &\quad + (\mathbf{d}\sigma_j)(d\alpha^j) + \sigma_j (dd\alpha^j) \\ &= (A_i^k \wedge A_j^i + dA_j^k)\sigma_k \wedge \alpha^j. \end{aligned}$$

Thus if we define $R_j^k = A_i^k \wedge A_j^i + dA_j^k$ then $\mathbf{dd}\alpha = R_j^k \sigma_k \wedge \alpha^j$. Since $R_j^k \sigma_k$ is a 2-form with values in $\xi[V]$, we think of R_j^k as the components of a 2-form R with values in $Hom(\xi[V], \xi[V])$. Thus we can write $\mathbf{dd}\alpha =$

$R \wedge \alpha$. It is clear that second covariant derivative does not generally vanish. We will see that R is intimately related to the curvature tensor, and is in fact equal to the local representative F described earlier. Proving this is only complicated by our use of compound indices in the vector bundle case. Firstly, we calculate the structure coefficients of $L(\mathrm{Aut}(V))$. This is achieved easily by matrix multiplication, obtaining $[\varepsilon^i_j, \varepsilon^k_l] = \varepsilon^i_l \delta^k_j - \varepsilon^k_j \delta^i_l$. Inserting the corresponding structure coefficients into our earlier expression for F, we obtain $F^k_j = R^k_j$. Thus the second exterior derivative not being zero is an expression of the fact that the curvature is not zero. It can further be shown, again in Darling [3], that if $s \in \Gamma\xi[V]$ then $(\mathbf{dd}s)(X,Y) = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s$. This relates R to the usual Riemann tensor of Riemannian geometry.

8. Summary

We have begun by motivating a series of abstract definitions by means of the familiar example of the tangent space of a manifold, and the idea of local frames. Principal fibres bundles have been the general setting in which connections, in three different formulations, are understood. These connections allow differentiation in an associated vector bundle. In particular, a connection on the frame bundle leads to a covariant derivative on the tangent bundle. Various exterior derivatives have also been defined, for forms on the total space or base space of a bundle, and these ideas have been used to introduce briefly the curvature of a connection, and the relationship with more familiar notions of curvature.

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